From *Teaching in the AI Era* by Muhammad Tayyab Ilyas (2026), https://doi.org/10.5281/zenodo.24085773. These pages may be copied for use in your own school.

Appendix C. Practical toolkit

These four templates gather routines from the chapters into drafts you can adapt. They are working drafts, not law and not adopted school policy. Nobody has tested them as a package, so no trial result carries over to them. Check each one against your school's existing rules, current age terms and local duties before use. Throughout, teacher-facing preparation stays separate from direct student use.

When resources are thin. Chapters 4, 7 and 11 describe schools, many of them in Pakistan, where one teacher's phone, a WhatsApp group and paper do most of the work. Every template here works on paper. Start with teacher-facing uses, which need no student device or account. Keep the class agreement and the family letter short enough to read aloud at a meeting. Have a competent speaker check any Urdu or home-language version before it goes out. These templates make no claim about Pakistani law or provincial rules.

Classroom AI policy (adaptable draft)

This draft covers one class or subject. It becomes a working agreement only when your school checks it against existing policies and adopts it through its own process. Complete each part in plain words, then read it through with the class. Chapters 1, 5, 8 and 11 supply the routines behind it.

1. **Purpose.** Say in two sentences what the agreement protects, such as students' own thinking and fair assessment.
2. **Who uses AI.** Separate teacher-facing preparation from direct student use. Record student use only for a tool whose current age terms and permissions the school has checked.
3. **Help allowed for each task.** For each major task, describe which assistance students may use and which they may not, before work begins. A brainstorm and an unaided check may need different rules.
4. **Saying what helped.** Explain how students note their assistance, for example what they asked and what they changed. Treat that note as information, not a confession.
5. **Personal information.** Keep names, photographs, marks and other identifying details out of any tool unless the school's data rules allow it.
6. **Tools.** List only tools the school has checked and approved. If it has no approval process yet, agree one before students use any tool. Name the person who keeps the list.

7. **A route without AI.** Offer a way to reach the same learning goal without a device or account. Keep private the reasons a student chooses it.
8. **Human review.** Name the adult who checks AI-assisted feedback, grouping or advice before it affects a student.
9. **When a concern arises.** Detector output, sudden fluency or hesitation prove nothing by themselves. Describe how the teacher will talk with the student, gather comparable evidence and follow the school's conduct procedure, or agree one first.
10. **Questions and review.** Name who answers questions from students, staff and families. Add an owner and a review date, and revisit the agreement each term.

Before adoption, compare the draft with your behaviour, assessment, data and safeguarding policies. TeachAI's sample guidance offers a longer comparison and calls itself illustrative and "not a definitive model" (TeachAI, 2025a). Its context is American, so treat it as a prompt for discussion.

Letter to families (adaptable draft)

Before you use this letter. This is a draft to adapt, not a message to send. Replace each italic instruction with checked local details, or delete it. Confirm every statement about tools, accounts, data and permission with the responsible colleague first. Brief staff before families, as Chapter 11 suggests. Ask a competent speaker to check any translation, and offer an oral explanation to families who prefer one (Chapters 7 and 8).

Dear families,

This letter explains how our class may use artificial intelligence (AI) tools this term. It also tells you how to ask questions.

We want students to keep doing their own thinking. They will also learn how these tools work and where they go wrong. *Describe one sample activity and what pupils will learn from it.*

Some uses are for teachers only, such as preparing practice questions. Those uses need no account for your child. *State whether students will use any tool directly. If they will, name it and confirm its current age terms with the responsible colleague.*

Add checked information about accounts, what data the tool keeps and who can see it. If you cannot confirm these details, do not send the letter yet.

A teacher will check AI-assisted material before students see it and will stay responsible for marks and feedback. *Keep this sentence only if it describes what will actually happen.*

Your child can reach the same learning goal without using an AI tool. You do not need to give a reason for choosing that route. *Describe the alternative activity. Keep these sentences only if the school really offers one.*

If you have a question or a worry, please talk to us privately. *Add a named contact and a time or way to reach them.*

If the school needs a decision from families, explain its actual permission process here and allow time for questions first. Otherwise delete this instruction.

With thanks,

Add the teacher's name and role.

TeachAI publishes a sample letter that you can compare with this one (TeachAI, 2025b). It names a product age rule that may since have changed, so check any age terms on the day. A missing reply never tells you that a family understood or agreed (Chapter 8).

Assessment redesign checklist

Use this before you set or revise a task where AI help could change what a result means. It draws on Chapters 5, 6 and 12. Work through the questions in order, alone or with a colleague.

1. **The claim.** What should a good result let you say about this learner?
2. **Help that changes the claim.** Which assistance leaves that claim intact, and which would undermine it? Tell students before work begins.
3. **Evidence from the learner.** Where will you see the thinking directly? Drafts in view, a short quiz with devices closed or a question about their own work can all help.
4. **Access.** Can every student show the learning in the formats on offer? Keep agreed access arrangements in protected work, and supply common materials where private tools could shape the result.

5. **Comparison before judgement.** Set each new piece beside earlier work. Agreement supports a broader claim, and a mismatch calls for another learning check.
6. **Concerns.** Detector output never settles misconduct by itself, and hesitation does not prove AI use. Record uncertainty honestly, and keep any conduct question in a separate process.
7. **Feedback.** Decide what the learner will do next with the result.
8. **Moderation.** Compare sample responses with colleagues before any mark that carries consequences.
9. **Staff time.** Count the hours the redesign needs. A rotating sample of conversations can guide teaching but cannot certify students who were not sampled.

Higher education connection. Chapter 12 applies the same questions to university modules. Its routines name the outcomes students must show without AI, calibrate examiners before oral checks, separate language fluency from knowing and budget the hours first. Perkins et al. (2024a) offer a published scale for setting permitted AI use by assessment task. Its authors designed it for higher education, so read it at the source and adapt with care.

Planning prompts

These cards suit teacher-facing preparation, which needs no student account. They come from Chapter 4's routines, which nobody has yet tested as a programme.

1. **Think first.** Before opening any tool, write the goal, the mistake you expect and how you will check learning.
2. **Context without names.** Describe the class by year group, reading level and prior learning. Leave out names and anything that could identify a student.
3. **Ask for a draft that fits.** Try wording such as ‘Suggest five practice questions on comparing fractions for Year 4. Include two that reveal whether students think a larger denominator means a larger fraction.’
4. **Ask for likely mistakes.** Another example is ‘List three common misconceptions about photosynthesis for 13-year-olds, with one quick check for each.’
5. **Refine, then keep.** Add follow-up requests until the draft fits your three lines. In one English trial, a teacher’s prompts grew as they added the year group and reading age. Another teacher refined outputs with follow-up requests (Roy et al., 2024, pp. 69, 75).
6. **Check before students see it.** Check every source, quotation and consequential fact against a trusted source. Chapter 4 asks for every source and at least one claim, and this card asks for more.
7. **Share what worked.** Save useful prompts in a department bank with a note on the class and topic. One trial teacher asked colleagues to share prompts that had worked (Roy et al., 2024, p. 75).

8. **Count honestly.** Time prompting, checking and editing for two weeks before deciding whether a tool saves time. A diary describes your experience, not a proven saving.
9. **Prepare a paper fallback.** Draft while the connection holds, then copy the questions onto the board or print one sheet.

Worksheets to copy

These pages turn the routines in the chapters into forms you can copy, print or draw on the board. Each one names the chapter it comes from. Write on them in pencil, keep the ones that help and drop the rest.

Worksheet 1. Decision log (Chapter 13)

Keep this where you plan. A few lines a week show who answered for each result, and they make the next conversation with a colleague or a parent easier.

Date	Task	What the tool drafted	What I changed	Why

Worksheet 2. Context card without names (Chapter 4)

Fill one card for each class you plan for with AI. Keep it beside you when you prompt, and keep every student's name off it.

Item	Your class
Year group or class	
Reading level, in a phrase	
What they already know about this topic	
The goal of this lesson	
The mistake I expect	
How I will check their thinking	
What I will do if the connection fails	

Worksheet 3. Order-of-work planner (Chapter 5)

One row per lesson. The student tries first, help comes next, and later the student shows without help what they can do.

Lesson or topic	Try first: what students attempt alone	Help next: what help, from whom	Show alone: how and when

Worksheet 4. What the mark lets me claim (Chapter 6)

Use before you set a piece of work that will carry a mark. If the help column would break the claim, change the task or the conditions, not the student.

Task	What a good result lets me claim	Help the task allows	Help that would break the claim	Where I will see the thinking

Worksheet 5. Who can actually use it? (Chapter 7)

Before any activity that needs a tool, list the groups in your class and tick what each one has. The last column is not optional.

Group of students	Device	Connection	Language	Time at home	Route without AI

Worksheet 6. Two-week time count (Chapters 4 and 10)

Time your prompting, checking and editing for ten working days before deciding whether a tool saves you anything. Minutes, not impressions.

Day	Prompting	Checking	Editing	Minutes the tool saved	Notes
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Total					

Worksheet 7. The four floors in my school (Chapters 1 and 13)

Fill this with colleagues, one floor at a time, starting at the bottom. The last column is where a staffroom request begins.

Floor	What we have	What is missing	Who can change it
Strategic practice (skilled choices about tools)			
Developmental opportunities (time and chances to learn)			
Psychological safety (safe to try and make mistakes)			
Structural provisions (power, devices, connection)			

Lesson cards from Chapters 7 and 9

These cards give the full lesson designs that Chapters 7 and 9 refer to. They are proposals for adaptation, not accounts of observed lessons, and no study has tested them as a programme.

Card 1. Primary reading with access in view (Chapter 7)

Primary reading with access in view. Consider a Year 4 lesson about how plants move water. The teacher wants students to explain the process using a diagram and a short paragraph. She keeps that goal common across the class. Before preparing another worksheet, she identifies what blocks participation for particular students.

Some students may understand the spoken explanation but struggle with the printed passage. She tries a checked read-aloud version for those students. Others may need the main ideas discussed in a familiar language before meeting the school's scientific vocabulary. A diagram remains available to everyone. These routes lead towards the same explanation.

If she uses AI to shorten the passage, she checks whether the shortened version preserves the movement of water through the plant. She compares it with the textbook. A shorter text that removes the causal link would defeat the lesson's purpose. Students should still explain how the parts relate.

During the activity, the teacher asks a student using read-aloud support to point to the relevant part of the diagram. The response helps her judge access to meaning. She checks decoding separately during reading instruction. Keeping those purposes distinct prevents support in one lesson from hiding a need in another.

She also asks whether the student finds the voice comfortable and whether replaying a sentence helps. The student can choose a paper version or adult reading if the tool distracts. She records a brief observation about participation and understanding. Opening the audio file alone would tell her very little.

Card 2. Secondary mathematics with shared control (Chapter 7)

Secondary mathematics with shared control. Consider a secondary class practising ratios after a teacher's explanation. A few approved devices offer an AI tutor during a supervised station. The teacher supplies a checked starting problem. She keeps equivalent paper practice available, so a failed connection does not stop the lesson.

Working in pairs, students take turns entering questions and explaining the tutor's reply. Changing roles prevents keyboard access from becoming one student's permanent privilege. The teacher watches who can formulate a useful question and who needs help to begin. Confident students should not absorb all the conversation.

A pair might ask why two ratios describe the same relationship. Before accepting the explanation, both students connect it to the diagram in their books. If the reply confuses them, the teacher intervenes. The point of supervision is to notice that confusion before another fluent answer buries it.

Later, she uses Chapter 5's check of understanding with the tutor closed. She compares the work of students starting at different attainment levels. A class average would miss a pattern in which confident students progress while beginners remain stuck. She also notes whether device difficulties consume some students' practice time.

The shared roles come from the Sierra Leone trial, which also supplied technical support and more devices than one classroom phone. This example borrows the idea, but it cannot

borrow the trial's results. Each school must judge for itself whether its own arrangement helps learning in its own conditions.

Card 3. Secondary writing across languages (Chapter 7)

Secondary writing across languages. An English teacher might allow students to clarify an instruction in a familiar language before planning a composition. Students then write the part that develops the intended English skill. The teacher explains the boundary in advance. Permission should be clear enough for every student to understand it.

For a student struggling with the task, she first checks whether the difficulty concerns the instruction, vocabulary or organisation of ideas. Each calls for different help. A translated instruction may solve the first. A conversation or model paragraph may suit the others. Asking a chatbot to complete everything would hide those distinctions.

She keeps translation checks small enough to manage. A bilingual colleague can review a reusable set of instructions or subject terms during planned collaboration time. The teacher checks any newly generated version again. Reusing approved language reduces repeated effort while leaving room to correct an unsuitable phrase.

Card 4. A support plan with the child present (Chapter 7)

A support plan with the child present. A teacher and special educational needs coordinator might use an approved tool to suggest clearer wording for a draft target. They begin from their

observations of the student. Identifying details remain outside the prompt. They then check whether the wording describes an observable and useful next step.

The student's preferred communication method shapes that conversation. Family knowledge may reveal why a proposed routine will fail at home or create unnecessary distress. A fluent draft must yield to that knowledge. Teachers keep responsibility for the goal and the support needed to work towards it.

Card 5. A family message that survives translation (Chapter 7)

A family message that survives translation. A school might prepare a short message about bringing materials for a science activity. Staff write the English clearly before requesting a translation. A competent speaker checks the action, date, cost and school terms. The school provides a way for families to ask questions without using the same app.

Staff should avoid sending a poorly checked translation merely because the English looks simple. A message concerning health, consent or an urgent change needs an appropriately reliable route. Chapter 8 develops those responsibilities. Here the equity point is that a family's language must not determine whether it receives an accurate instruction.

Card 6. Review the students who disappear (Chapter 7)

Review the students who disappear. After a trial, the teacher looks for students whose use or learning remains low. She checks absences, device turns, access needs and the suitability

of explanations. A short discussion with those students may reveal the next adjustment. Continued failure can justify stopping the tool and preserving the teaching through another route.

The review should include costs to staff and families. Extra preparation, borrowed devices and repeated troubleshooting belong beside any apparent saving. Teachers should not carry those costs silently in the hope that inclusion will eventually become easier. A workable next step may be smaller than the school's original ambition.

Card 7. Primary patterns on cards (Chapter 9)

Primary patterns on cards. A Grade III teacher could begin with drawings of leaves showing different shapes and colours. The teacher chooses familiar categories and prepares examples for each. Students inspect the labelled cards and describe the patterns they notice. Nobody supplies photographs of children or other personal information.

Next, the teacher brings out unfamiliar cards. Some fit the earlier examples easily, while others combine features in a different way. Pairs predict a category and explain what makes them uncertain. The teacher records disagreement without treating a quick answer as the aim.

Together, the class considers what would happen if all the training examples showed only one kind of leaf. Students suggest which missing examples would improve the collection. Their task concerns the evidence available to a system. The teacher keeps that purpose distinct from the children's own botanical knowledge.

A permitted teacher-led model demonstration could follow when equipment allows it. Students compare their predictions with the model's outputs and look for patterns in its mistakes. Without equipment, the teacher explains which part remains a paper representation. No claim follows that students have trained real software.

To close, each student draws an example that the original collection failed to represent. They explain why adding it might help. The teacher listens for a link between data and errors. Sorting correctly matters less here than explaining the missing evidence.

Card 8. Later primary answers worth checking (Chapter 9)

Later primary answers worth checking. A teacher could prepare a short generated paragraph about a topic the class has already studied. Before the lesson, the teacher checks every factual claim and keeps the source passages. The paragraph may contain an error, but the teacher never depends on the tool making one. Any deliberate alteration needs a clear explanation afterwards.

Students first decide which claim they can check using their class book. They mark where the book supports, contradicts or leaves the claim unresolved. Partners explain their decision using the relevant passage. The teacher includes a correct claim so that rejection alone earns no reward.

A student who cannot yet read the passage independently can hear it read aloud. Their explanation can also be oral. The learning concerns the relationship between a claim and evidence. Reading difficulty should not silently decide who appears competent about AI.

At the end, the teacher identifies the generated material and any deliberate changes. Students revisit their first judgement. They explain what changed their mind, or why they kept their original view.

Card 9. Lower-secondary questions with a purpose (Chapter 9)

Lower-secondary questions with a purpose. A science teacher could use a prepared chatbot exchange about an idea the class knows. Students identify what the original prompt leaves unclear. They draft a follow-up question and explain what a useful reply would need to contain. Where direct use is permitted, the teacher can test selected questions.

The next step is external checking. Students compare a claim from the reply with an appropriate science source or classroom observation. They record whether the follow-up improved clarity and whether the claim gained support. Those are separate judgements. A clearer answer can still be wrong.

Clerc's workshop suggests that practising follow-ups can change immediate behaviour. It does not establish that this particular lesson will do so. A later lesson can quietly present another incomplete exchange. The teacher watches whether students notice the gap without a reminder.

Card 10. Secondary testing that can fail usefully (Chapter 9)

Secondary testing that can fail usefully. A computing teacher could propose a classroom model for sorting pictures of recyclable objects. Students first define the categories and identify examples that might cause disagreement. They use approved materials that contain no personal information. The teacher checks the software and its account requirements beforehand.

Students divide their pictures into a training collection and a separate testing collection. They explain why testing only familiar examples would tell them too little. When the model makes an error, they record the object and the conditions. A wrong result becomes evidence for their next decision.

The teacher can then introduce an unfamiliar background or a damaged object. Students predict whether the change will affect the result. They compare those predictions with the model's response and revise their explanations. Simply adding more pictures should not end the discussion. Their choice of pictures needs a reason.

The class also considers the cost of different mistakes. Sending a harmless item to the wrong classroom pile differs from sorting dangerous waste incorrectly. Students identify where an adult would need to intervene. Nobody should use the classroom model to make real waste-handling decisions.

A student's final explanation names the intended use, the cases tested and the limits still unresolved. Those limits may justify abandoning the model. With no suitable software, the class can

inspect a teacher-prepared testing record instead. That version teaches evaluation of results while leaving actual model training for another occasion.

Card 11. Upper-secondary creation with a record of choices (Chapter 9)

Upper-secondary creation with a record of choices. A language teacher could ask students to produce an information sheet from a teacher-selected source passage. Permitted AI use might include suggesting an outline or proposing a shorter version. Students identify what they accepted, changed and rejected. They then check the sheet against the original passage.

The audience should affect those decisions. A version for younger readers may need simpler language without losing the source's uncertainty. Students explain where a proposed simplification would change the meaning. The teacher assesses that explanation alongside the finished sheet.

Anyone unable or unwilling to use AI can work from a prepared output supplied by the teacher. They make the same choices and complete the same comparison. A short discussion can ask which part demanded the student's own judgement. This preserves participation without requiring an account.

Card 12. A short check after the lesson (Chapter 9)

A short check after the lesson. Later, the teacher could supply a fresh claim with a convincing explanation. Students decide what they would check and where they would look. They should also explain what evidence would make them accept it. That final question guards against teaching blanket distrust.

The teacher can compare these responses with earlier work, using the same simple criteria. Can students explain the system, identify a checking action and justify a decision? Improvement in one part need not mean improvement in all. The next lesson can address the weakest part.